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Phase Distortion Statistics as a Representation of the Glottal Source: Application to the Classification of Voice Qualities

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Adaptive Harmonic Model

$$s_i(t) = \sum_{h=1}^{H_i} a_{i,h} \cdot e^{j(h\phi_0(t)+\phi_{i,h})} \qquad [1,2]$$

$$\phi_0(t) = \frac{2\pi}{f_s}\int_{t_i}^{t} f_0(\tau)d\tau$$

$f_0(t)$ Fundamental frequency (e.g. STRAIGHT, SWIPEP)

$a_{i,h}$ Instantaneous amplitude (e.g. MFCC, MGC, LSF) $A_i(f)$: linear interpolation of $a_{i,h}$

$\phi_{i,h}$ Instantaneous phase How to model it statistically (?!)

$H_i = \lfloor 0.5 f_s / f_0(t_i) \rfloor$ Number of harmonics (up to Nyquist!)

Properties

- Almost perfect resynthesis quality
- Full-band
- No voicing decision

Phase Distortion

Theoretical phase model

$$\phi_{i,h} = \overbrace{\underbrace{\theta_{i,h}}_{\text{source shape}} + \underbrace{h\frac{2\pi}{f_s}\int_{c_i}^{t_i} f_0(\tau)d\tau}_{\text{linear phase}}}^{\text{voice source}} + \overbrace{\angle A_i(hf_0(t_i))}^{\text{filter}}$$

Phase Distortion (PD) [6,7]

$$\widetilde{\phi}_{i,h} = \phi_{i,h} - \angle A_i(hf_0(t_i))$$

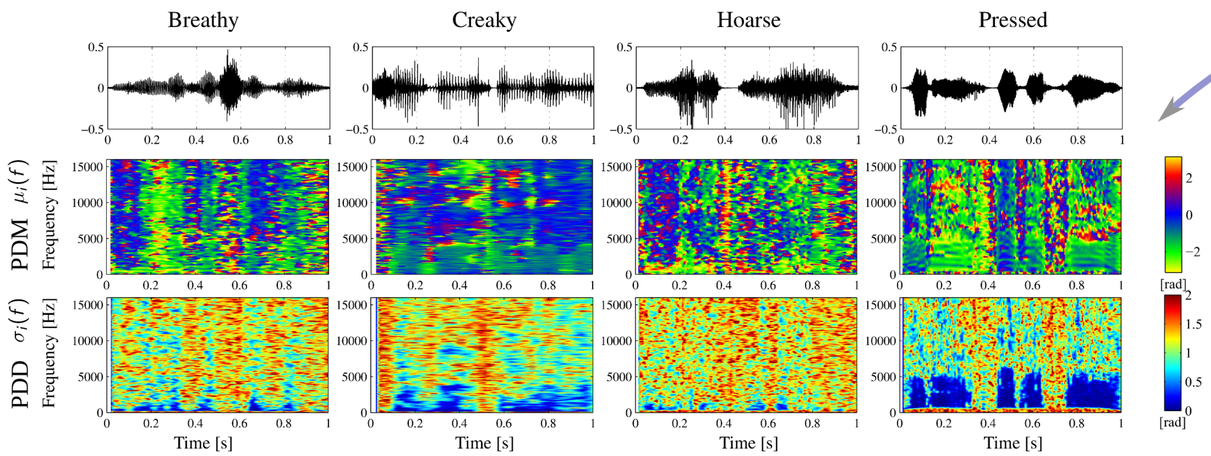
$$\widetilde{\mathrm{RPS}}_{i,h} = \widetilde{\phi}_{i,h} - h\widetilde{\phi}_{i,h} \qquad [8]$$

$$\mathrm{PD}_{i,h} = \underset{h}{\Delta}\widetilde{\mathrm{RPS}}_{i,h} = \widetilde{\mathrm{RPS}}_{i,h+1} - \widetilde{\mathrm{RPS}}_{i,h} = \theta_{i,h+1} - \theta_{i,h} - \theta_{i,1}$$

Properties

- Highly correlated to the shape of the voice source [3,4]

Features extraction on four voice qualities



Results of classification of voice qualities

(feature size in exponent)	BREATHY	CREAKY	HOARSE	PRESSED	TOTAL
MFCC^{13}	76.2	73.6	82.9	77.4	77.5
RD^{2}	66.3	67.2	64.3	66.5	**66.0**
$\Delta\,\mathrm{GCI}^{1}$	70.0	71.0	75.1	67.4	**71.0**
PDM^{25}	**77.9**	71.6	81.2	**79.2**	**77.5**
PDD^{25}	**75.0**	72.0	82.3	**76.0**	**76.5**
$\mathrm{MFCC}^{4}+\mathrm{RD}^{2}$	77.7	75.2	83.5	77.8	78.6
$\mathrm{MFCC}^{4}+\Delta\,\mathrm{GCI}^{1}$	77.4	74.8	85.0	76.7	78.4
$\mathrm{MFCC}^{4}+\mathrm{RD}^{2}+\Delta\,\mathrm{GCI}^{1}$	77.8	75.2	84.6	77.9	**78.9**
$\mathrm{MFCC}^{4}+\mathrm{PDM}^{25}$	78.7	75.7	84.7	81.1	80.0
$\mathrm{MFCC}^{4}+\mathrm{PDD}^{25}$	79.2	76.3	85.1	79.1	79.9
$\mathrm{MFCC}^{4}+\mathrm{PDM}^{25}+\mathrm{PDD}^{25}$	80.2	76.7	85.4	81.8	**81.0**

Database [5]

- 4000 speech recordings
- Expressive speech (video-game: Mass effect 3, French version)

Classification

- Acoustic representation : super-vectors (8-2048)
  i-vectors (10-400)
- Classification system : SVM (gaussian kernel)

Performance

- 10-fold cross-validation
- Balanced accuracy

Main results

- Representation of phase of the voice source, without analytical model or GCI
- Better than analytical models
- Complete the information carried in MFCC

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A Measure of Phase Randomness for the Harmonic Model in Speech Synthesis

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PD's short-term moments

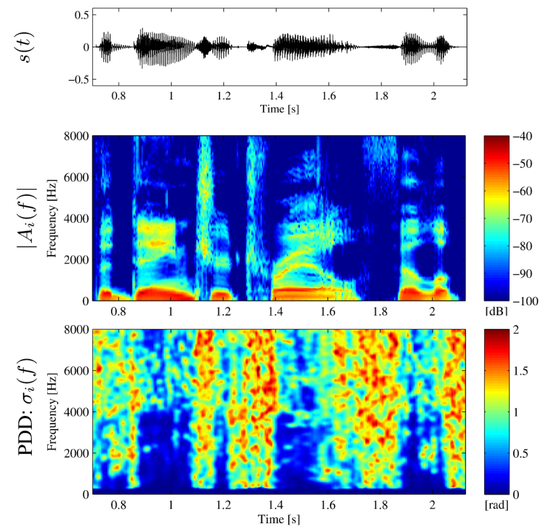
$\mathrm{PD}_{i,h} \Rightarrow \mathrm{PD}_i(f)$

PDM $$\mu_i(f) = \angle\left(\frac{1}{M}\sum_{m\in B} e^{j\mathrm{PD}_m(f)}\right)$$

where $B = i - \frac{M-1}{2} \cdots i + \frac{M-1}{2}$, and $M = 9$.

PDD $$\sigma_i(f) = \sqrt{-2\log\left|\frac{1}{M}\sum_{m\in C} e^{j\mathrm{PD}_m(f)}\right|}$$

Example of HMPD's features



HMPD Synthesis

Signal synthesis

$$\hat{s}(t) = \sum_{h=1}^{H} \hat{a}_h(t)\cdot\cos(\hat{\phi}_h(t))\cdot\chi_{[hf_0(t)<0.5f_s]}(t)$$

Amplitude synthesis

$\hat{a}_h(t)$: linear interpolation of $\hat{a}_{i,h}$ $\qquad \hat{a}_{i,h} = A_i(hf_0(t_i))$

Phase synthesis

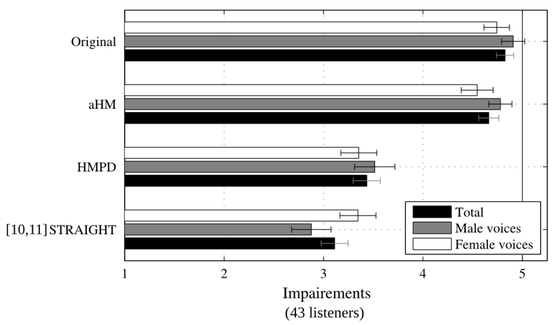
$$\hat{\mathrm{PD}}_{i,h} = \mathcal{WN}(0, \sigma_i(hf_0(t_i))) \qquad \mathcal{WN}\text{: Wrapped-Normal distribution [9]}$$

$$\hat{\mathrm{RPS}}_{i,h} = \underset{h}{\Delta}^{-1}\hat{\mathrm{PD}}_{i,h} + \angle A_i(hf_0(t_i))$$

$$\hat{\phi}_h(t) = \hat{\mathrm{RPS}}_h(t) + h\frac{2\pi}{f_s}\int_0^t f_0(\tau)d\tau$$

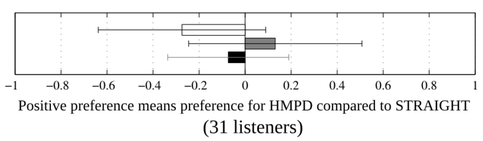
Results of HMPD's quality

Analysis/Re-synthesis



HMM-based synthesis [12]

- 2 English and 2 Spanish voices
- HTS v2.1.1
- 3 **Continuous** streams: [13,14]
  logf0(1), A-MFCC(39), PDD-MFCC(12)



Main results

- HMPD analysis/re-synthesis slightly better than STRAIGHT
- HMM-based synthesis similar between HMPD and STRAIGHT
- HMPD HMM-based synthesis without voicing decision (!)

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Source-code available on COVAREP
http://covarep.github.io/covarep